

Madanapalle, India

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github.com/hacker1514

pypi.org/project/kshell

hacker1514.github.io/make_it/main/

CORE ENGINEERING SKILLS

Systems & Architecture

32+ languages studied & practiced · C/C++ · Python · Java · JavaScript · Shell · Linux Kernel · OS Architecture · Dev Tooling

AI & ML Engineering

LLM Optimization · Mobile AI Inference · Resource-Constrained Model Execution · Local AI Infra · Browser Model Storage

Cybersecurity & Recon

Ethical Hacking · Linux & Network Security · Web/Android Security · Security Automation · Pentesting Concepts

Distributed Systems

Decentralized Mesh Networks · P2P Protocols · Server Administration · Android Memory Virtualization

KEY STRENGTHS

- Systems & algorithmic thinking
- On-device AI optimization
- Rapid technical prototyping
- Independent research rigor
- Cross-domain engineering
- Cybersecurity mindset
- Linux systems expertise

NIRANJAN KUMAR K

Systems & AI Software Engineer · Cybersecurity Researcher

Systems-focused software and AI engineer with strong practical experience across operating system architecture, artificial intelligence, cybersecurity research, Linux systems, and mobile computing. Creator of **KNI OS** and **K Programming**, with published Python packages including **KShell** on PyPI. Passionate about solving complex engineering challenges involving AI execution on constrained hardware, mobile LLM memory optimization, local model storage primitives, and distributed networking systems.

SOFTWARE & AI SYSTEMS PROJECTS

KShell — Developer Tooling & Automation Package | **PyPI Package**

Creator and developer of KShell, an open-source Python package published on PyPI for shell automation and developer productivity. (**pypi.org/project/kshell/**)

KNI / KNI OS & K Programming Ecosystem | **OS Architecture**

Creator of the KNI technology ecosystem, focusing on operating-system design primitives, compiler/interpreter semantics, and low-level system software tooling.

Mobile LLM Optimization & Edge AI Inference | **AI Research**

Research and experimentation focused on executing large AI models efficiently on mobile hardware through algorithmic weight compression, memory pooling, and browser-based model caching.

Decentralized Mesh & Storage Architecture Concepts

- **Global Connectivity Concept:** Experimental networking architectures designed to connect systems via mesh/P2P approaches without centralized Internet dependencies.
- **Distributed Android Storage:** Prototyped concepts for extending Android device storage using external or distributed memory resources.

SELECTED ACHIEVEMENTS & RESEARCH

- **PyPI Package Publisher:** Created and distributed open-source Python packages used in developer workflows (e.g. `kshell`).
- **Platform & Security Reconnaissance:** Identified and analyzed **52+ web platforms and technology services** as part of independent security research and architectural reconnaissance.
- **Edge AI Optimization:** Prototyped lightweight execution primitives enabling LLM inference on resource-constrained mobile hardware and browser environments.

EDUCATION

Bachelor of Engineering / Technology — Madanapalle Institute of Technology & Science (MITS), Madanapalle

Intermediate Education — Sri Chaitanya Junior College, Tirupati

Secondary Education — Z.P. High School, Chembakur

ENGINEERING PHILOSOPHY & OBJECTIVE

> *"Build what meems difficult. Understand how it works. Optimize it until it becomes practical."*

Driven to contribute to Google-scale software infrastructure, next-generation AI systems, and low-level operating system architectures that impact billions of users worldwide.